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The Equational Theories Project: Advancing Collaborative Mathematical Research at Scale

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Abstract. We report on the *Equational Theories Project* (ETP), an online collaborative pilot project to explore new ways to collaborate in mathematics with machine assistance. The project successfully determined all 22 028 942 edges of the implication graph between the 4694 simplest equational laws on magmas, by a combination of human-generated and automated proofs, all validated by the formal proof assistant language *Lean*. As a result of this project, several new constructions of magmas satisfying specific laws were discovered, and several auxiliary questions were also addressed, such as the effect of restricting attention to finite magmas.

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1. Introduction

The purpose of this paper is to report on the *Equational Theories Project* (ETP)[1], a pilot project launched [65] in September 2024 to explore new ways to collaboratively work on mathematical research projects using machine assistance. The project goal, in the area of universal algebra, was selected[2] to be particularly amenable to crowdsourced and computer-assisted techniques, while still being of mathematical research interest.

The project achieved its primary goal on 14 April 2025, when the $4694 \times (4694 - 1) = 22\,028\,942$ implications between the test set of 4694 equational laws were completely determined, with proofs or refutations formalized in *Lean*. This required coordinating the efforts of a large number of participants contributing both human-written formalizations and automatically generated proofs from various computer tools. In this paper, we report on both the scientific outcomes of the project, as well as the organizational issues that came up with organizing a mathematical project of this scale.

1.1. Magmas and equational laws.

In order to describe the mathematical goals of the ETP, we need some notation. A *magma* $\mathcal{M} = (M, \diamond)$ is a set M (known as the *carrier*) together with a binary operation $\diamond\colon M \times M \to M$. An *equational law* for a magma, or *law* for short, is an identity involving $\diamond$ and some formal indeterminates, which we will typically denote using the Roman letters $\mathrm{x}, \mathrm{y}, \mathrm{z}, \mathrm{w}, \mathrm{u}, \mathrm{v}$, as well as the formal equality symbol $\simeq$ in place of the equality symbol $=$ to emphasize the formal nature of the law. If M is finite, we refer to its cardinality as the *size* of the magma $\mathcal{M}$.

An *equational theory* is a collection of equational laws; in this paper we will primarily be concerned with theories generated by a single such law, although it is certainly of interest to explore larger theories as well. Equational theories are one of the simplest non-trivial examples of a theory in the model-theoretic sense; they also arise in various areas of computer science, such as term rewriting systems [7], automated theorem proving [45], and in the Dolev–Yao model [23] of interactive cryptographic protocols.

In the ETP, a unique number was assigned to each equational law, via a numbering system that we describe in Appendix A. For instance, the *commutative law* $\mathrm{x} \diamond \mathrm{y} \simeq \mathrm{y} \diamond \mathrm{x}$ is assigned to the equation number E43, while the *associative law* $\mathrm{x} \diamond (\mathrm{y} \diamond \mathrm{z}) \simeq (\mathrm{x} \diamond \mathrm{y}) \diamond \mathrm{z}$ is assigned to the equation E4512. A list of all equations referred to by number in this paper is also provided in Appendix A.

A magma $\mathcal{M} = (M, \diamond)$ satisfies a law E if the law E holds for all possible assignments of the indeterminates to elements of M, in which case we write $\mathcal{M} \models \mathrm{E}$. Thus, for instance $\mathcal{M} \models \mathrm{E43}$ if one has $x \diamond y = y \diamond x$ for all $x, y \in M$. Note that the formal indeterminate symbols x, y in E43 are now replaced by concrete elements x, y of the carrier M.

We say that a law E *entails* or *implies* another law E′ if every magma that satisfies E, also satisfies E′: $(\mathcal{M} \models \mathrm{E}) \implies (\mathcal{M} \models \mathrm{E}')$. We write this relation as $\mathrm{E} \models \mathrm{E}'$. We say that two

[1] `https://teorth.github.io/equational_theories/`

[2] The specific mathematical goal was inspired by the MathOverflow question "Is there an identity between the associative identity and the constant identity?", posed on 17 July 2023.

laws are *equivalent* if they entail each other. For instance, the constant law $x \diamond y \simeq z \diamond w$ (E46) can easily be seen to be equivalent to the law $x \diamond x \simeq y \diamond z$ (E41). It is clear that $\models$ is a pre-order, that is to say a partial order after one quotients by equivalence.

In this entailment pre-ordering, the maximal element is given by the trivial law $x \simeq x$ (E1), and the minimal element is given by the singleton law $x \simeq y$ (E2), thus E2 $\models$ E $\models$ E1 for all laws E.

We also define a variant: we say that E *entails* E′ *for finite magmas*, and write $E \models_{\text{fin}} E'$, if every *finite* magma that satisfies E, also satisfies E′. Clearly, the relation $E \models E'$ implies $E \models_{\text{fin}} E'$; but, as observed by Austin [6], the converse is not true in general.

The *order* of an equational law is the number of occurrences of the magma operation, and can be viewed as a crude measure of complexity of the law. For instance, the commutative law E43 has order 2, while the associative law E4512 has order 4. We note some selected laws of small order that have previously appeared in the literature:

- The *central groupoid law* $x \simeq (y \diamond x) \diamond (x \diamond z)$ (E168) is an order-3 law introduced by Evans [26] and studied further by Knuth [35] and many further authors, being closely related to central digraphs (also known as unique path property digraphs), and leading in particular to the discovery of the Knuth-Bendix algorithm [36]; see [40] for a more recent survey.
- *Tarski's axiom* $x \simeq y \diamond (z \diamond (x \diamond (y \diamond z)))$ (E543) is an order-4 law that was shown by Tarski [66] to characterize the operation of subtraction in an abelian group; that is to say, a magma $\mathcal{M} = (M, \diamond)$ satisfies E543 if and only if there is an abelian group structure on $\mathcal{M}$ for which $x \diamond y = x - y$ for all $x, y \in M$.
- In a similar vein, it was shown in [50] (see also [51]) that the order-4 law $x \simeq (y \diamond z) \diamond (y \diamond (x \diamond z))$ (E1571) characterizes addition (or subtraction) in an abelian group of exponent 2; it was shown in [47] that the order-6 law $x \simeq (y \diamond ((x \diamond y) \diamond y)) \diamond (x \diamond (z \diamond y))$ (E345169) characterizes the Sheffer stroke in a boolean algebra, and it was shown in [29] that the order-8 law $x \simeq y \diamond ((((y \diamond y) \diamond x) \diamond z) \diamond (((y \diamond y) \diamond y) \diamond z))$ (E42323216) characterizes division in a (not necessarily abelian) group.

Some further examples of laws characterizing well-known algebraic structures are listed in [45].

The Birkhoff completeness theorem [7, Th. 3.5.14] implies that an implication $E \models E'$ of equational laws holds if and only if the left-hand side of E′ can be transformed into the right-hand side by a finite number of substitution rewrites using the law E. However, the problem of determining whether such an implication holds is undecidable in general [49]. Even when the order is small, some implications[3] can require lengthy computer-assisted proofs; for instance, it was noted in [34] that the order-4 law $x \simeq (y \diamond x) \diamond ((x \diamond z) \diamond z)$ (E1689) was equivalent to the singleton law $x \simeq y$ (E2), but all known proofs were found with computer assistance.[4] Furthermore, for the finite magma implication relation $E \models_{\text{fin}} E'$, no analogue of the Birkhoff completeness theorem is available.

[3] Another contemporaneous example of this phenomenon was the solution of the Robbins problem [44].

[4] We improved such a proof to make it human-readable, see the blueprint of the ETP.